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Cathode for all-solid-state battery for preventing short-circuiting between electrodes and method of manufacturing the same

Abstract

A method of manufacturing a cathode for an all-solid-state battery includes: stacking a protective member on a cathode current collector, the protective member including a protective layer and a mask layer disposed on the protective layer and having a central portion which is an empty space; coating the protective member with a cathode material so that the central portion of the protective member is filled with the cathode material; and removing the mask layer to form a cathode coating layer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0115626	12/2018	Sadana	N/A	H01M 4/131
2019/0288277	12/2018	Michibata	N/A	H01M 4/362
2020/0083525	12/2019	Lee	N/A	H01M 4/62

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106981684	12/2016	CN	N/A
108701843	12/2017	CN	N/A
109802165	12/2018	CN	N/A
2309581	12/2010	EP	N/A
2011-216403	12/2010	JP	N/A
2014-203740	12/2013	JP	N/A
6394480	12/2017	JP	N/A
101846748	12/2017	KR	H01M 4/0404

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims the benefit of priority to Korean Patent Application No. 10-2019-0114022, filed on Sep. 17, 2019 in the Korean Intellectual Property Office, the entire content of which is incorporated herein for all purposes by this reference.

TECHNICAL FIELD

(2) The present disclosure relates to a cathode for an all-solid-state battery and a method of manufacturing the same, and more particularly to, a cathode for an all-solid-state battery, which is capable of preventing short-circuiting between electrodes by applying an insulating layer having

insulating properties on the cathode, and a method of manufacturing the same.

BACKGROUND

(3) A secondary battery is a battery that repeats charging and discharging by bidirectionally converting chemical energy and electrical energy through oxidation/reduction chemical reactions, and generally includes four basic elements, that is, a cathode, an anode, a separator, and an electrolyte. At the electrodes, including the cathode and the anode, the material that actually causes a reaction among the electrode material components is referred to as an active material.

(4) For a lithium ion secondary battery, a liquid electrolyte and an electrolyte including a liquid are used. However, the liquid electrolyte is disadvantageous because of volatility, a risk of explosion, and low thermal stability. In contrast, an all-solid-state battery using a solid electrolyte has a low risk of explosion and superior thermal stability. Moreover, the electrodes are stacked using a bipolar plate to enable series connection, resulting in high operating voltage. Hence, higher energy density is expected compared to cells that are connected in parallel using a liquid electrolyte.

(5) Electrode-electrolyte contact problems greatly occur in all-solid-state batteries, unlike conventional batteries using liquid electrolytes. Since the all-solid-state battery transfers lithium ions through solid-solid contact, an additional pressurization or heat treatment process between the electrode and the electrolyte is essential in order to improve contact between the electrode and the electrolyte upon cell stacking. As the pressure that is applied is larger, the contact between the electrode and the electrolyte is improved, thereby enhancing cell characteristics. In this case, however, the inside of the cell is more frequently shorted. Moreover, in the case of a cell stack rather than a single cell, short-circuiting of the cell more frequently occurs on account of non-uniform stress due to area mismatch upon electrode stacking.

(6) The information included in this Background section is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.

SUMMARY

(7) An objective of the present disclosure is to solve a short-circuiting problem during the manufacture of an electrode.

(8) Another objective of the present disclosure is to provide a structure that is able to reduce electrode short-circuiting and a defect rate.

(9) Still another objective of the present disclosure is to improve processability by enabling continuous processing when manufacturing an electrode for an all-solid-state battery.

(10) The objectives of the present disclosure are not limited to the foregoing, and will be able to be clearly understood through the following description and to be realized by the means described in the claims and combinations thereof.

(11) According to an exemplary embodiment of the present disclosure, a method of manufacturing a cathode for an all-solid-state battery includes: preparing a protective member including a protective layer and a mask layer disposed on the protective layer, the protective member having therein a central portion which is an empty space; stacking a protective member on a cathode current collector, the protective member including a protective layer and a mask layer disposed on the protective layer and having a central portion which is an empty space coating the protective member with a cathode material so that the central portion of the protective member is filled with the cathode material; and removing the mask layer to form a cathode coating layer.

(12) The mask layer may include polyethylene naphthalate (PEN), and the protective layer may include polyethylene naphthalate (PEN).

(13) In the preparing of the protective member, the mask layer may be stacked on the upper surface of the protective layer and a release sheet may be attached to the lower surface of the protective layer.

(14) An adhesive may be interposed between the protective layer and the mask layer.

- (15) In the stacking of the protective member, an adhesive layer may be interposed between the cathode current collector and the protective member, and the adhesive layer may include any one selected from the group consisting of ethylene vinyl acetate (EVA) copolymer, polyvinyl acetate (PVA), polyethylene (PE), amorphous polypropylene, thermoplastic elastomer, polyamide, polyester and combinations thereof.
- (16) The stacking of the protective member on the cathode current collector may be performed using either or both of heat and pressure.
- (17) In the removing of the mask layer, the mask layer and the cathode material applied on the mask layer may be removed, and the cathode material may remain in the form of being inserted into the central portion of the protective layer to form the cathode coating layer.
- (18) The thickness L3 of the cathode material may be greater than or equal to a sum of the thickness L1 of the protective layer and the thickness L2 of the mask layer.
- (19) The thickness L3' of the cathode coating layer may be greater than or equal to the thickness L1 of the protective layer.
- (20) The method of the present disclosure may further include punching the protective layer, the cathode coating layer and the cathode current collector after removing the mask layer.
- (21) The punching of the protective layer, the cathode coating layer and the cathode current collector may be performed at a predetermined interval apart from the central portion of the protective layer, and the upper surface and the lower surface of the punched protective layer may have any one shape of a circular shape and a polygonal shape.
- (22) According to another exemplary embodiment of the present disclosure, a cathode for an all-solid-state battery includes a cathode current collector, a cathode coating layer folioed on the cathode current collector, and an insulating layer disposed on the cathode current collector and having therein a central portion, which is an empty space, in which the upper surface of the central portion has any one shape of a circular shape and a polygonal shape, and the cathode coating layer is inserted into the central portion so that a portion of an outer side surface of the cathode coating layer and a side surface of the central portion are in contact with each other.
- (23) The cathode current collector may include any one selected from the group consisting of nickel, copper, zinc, aluminum and combinations thereof.
- (24) The cathode coating layer may include an active material and a binder.
- (25) The insulating layer may include an insulator.
- (26) The insulating layer may include polyethylene naphthalate (PEN).
- (27) The thickness of the insulating layer may be 0.80 to 1.30 times the thickness of the cathode coating layer.
- (28) The insulating layer may have a thickness of 100 μm or less and the cathode coating layer may have a thickness of 80 to 115 μm .
- (29) The inner diameter $\varnothing_{\text{sub.x}}$ of the insulating layer may be equal to the diameter $\varnothing_{\text{sub.z}}$ of the cathode coating layer, and the outer diameter $\varnothing_{\text{sub.y}}$ of the insulating layer may satisfy Expression 1 below.
- 1.1 $\varnothing_{\text{sub.x}} \leq \varnothing_{\text{sub.y}} \leq 2.0\varnothing_{\text{sub.x}}$ [Expression 1]
- (30) In addition, the present disclosure provides an all-solid-state battery, including the cathode described above, an anode, and a solid electrolyte interposed between the cathode and the anode.
- (31) According to the present disclosure, a problem of short-circuiting when manufacturing an electrode can be solved.
- (32) According to the present disclosure, a novel structure that is able to reduce electrode short-circuiting and a defect rate can be provided.
- (33) According to the present disclosure, processability can be increased by enabling continuous processing when manufacturing an electrode for an all-solid-state battery, and economic benefits can be obtained.

(34) The effects of the present disclosure are not limited to the foregoing, and should be understood to include all effects that can be reasonably anticipated from the following description.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 schematically shows a process of manufacturing a cathode for an all-solid-state battery according to the present disclosure;
- (2) FIG. 2 shows thicknesses of the components included in the cathode for an all-solid-state battery according to the present disclosure;
- (3) FIG. 3 shows a side view and a top plan view of the cathode for an all-solid-state battery according to the present disclosure before and after a punching process;
- (4) FIGS. 4A, 4B, and 4C show diameters of an insulating layer, a cathode coating layer, an anode and a solid electrolyte according to the present disclosure;
- (5) FIG. 5 shows scanning electron microscope (SEM) images of the cross-section of the all-solid-state battery manufactured in Example 1 according to the present disclosure;
- (6) FIG. 6 shows a charge/discharge curve of the all-solid-state battery manufactured in Example 1 according to the present disclosure; and
- (7) FIG. 7 shows the results of charging in the case of Comparative Example 1.

DESCRIPTION OF SPECIFIC EMBODIMENTS

(8) The above and other objectives, features and advantages of the present disclosure will be more clearly understood from the following preferred embodiments taken in conjunction with the accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed herein, and may be modified into different forms. These embodiments are provided to thoroughly explain the disclosure and to sufficiently transfer the spirit of the present disclosure to those skilled in the art.

(9) Throughout the drawings, the same reference numerals will refer to the same or like elements. For the sake of clarity of the present disclosure, the dimensions of structures are depicted as being larger than the actual sizes thereof. It will be understood that, although terms such as “first”, “second”, etc. may be used herein to describe various elements, these elements are not to be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a “first” element discussed below could be termed a “second” element without departing from the scope of the present disclosure. Similarly, the “second” element could also be termed a “first” element. As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(10) It will be further understood that the terms “comprise”, “include”, “have”, etc., when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof. Also, it will be understood that when an element such as a layer, film, area, or sheet is referred to as being “on” another element, it can be directly on the other element, or intervening elements may be present therebetween. Similarly, when an element such as a layer, film, area, or sheet is referred to as being “under” another element, it can be directly under the other element, or intervening elements may be present therebetween.

(11) Unless otherwise specified, all numbers, values, and/or representations that express the amounts of components, reaction conditions, polymer compositions, and mixtures used herein are to be taken as approximations including various uncertainties affecting the measurements that essentially occur in obtaining these values, among others, and thus should be understood to be modified by the term “about” in all cases. Furthermore, when a numerical range is disclosed in this

specification, the range is continuous, and includes all values from the minimum value of said range to the maximum value thereof, unless otherwise indicated. Moreover, when such a range pertains to integer values, all integers including the minimum value to the maximum value are included, unless otherwise indicated.

(12) In the present specification, when a range is described for a variable, it will be understood that the variable includes all values including the end points described within the stated range. For example, the range of “5 to 10” will be understood to include any subranges, such as 6 to 10, 7 to 10, 6 to 9, 7 to 9, and the like, as well as individual values of 5, 6, 7, 8, 9 and 10, and will also be understood to include any value between valid integers within the stated range, such as 5.5, 6.5, 7.5, 5.5 to 8.5, 6.5 to 9, and the like. Further, for example, the range of “10% to 30%” will be understood to include any subranges, such as 10% to 15%, 12% to 18%, 20% to 30%, etc., as well as all integers including values of 10%, 11%, 12%, 13% and the like up to 30%, and will also be understood to include any value between valid integers within the stated range, such as 10.5%, 15.5%, 25.5%, and the like.

(13) The present disclosure pertains to a method of manufacturing a cathode for an all-solid-state battery and a cathode for an all-solid-state battery manufactured by the method. The method of manufacturing the cathode for an all-solid-state battery, which is a method disclosure, and the cathode for an all-solid-state battery, which is a product disclosure, are described below.

(14) Here, a description of overlapping features in the method disclosure and the product invention will be omitted.

(15) Method of Manufacturing Cathode for all-Solid-State Battery

(16) The method of manufacturing the cathode for an all-solid-state battery according to the present disclosure includes preparing a protective member including a protective layer and a mask layer disposed on the protective layer and having therein multiple central portions, which are empty spaces, stacking the protective member on a cathode current collector, coating the protective member with a cathode material so that the central portions of the protective member are filled with the cathode material, and removing the mask layer to form a cathode coating layer.

(17) FIG. 1 shows the steps for manufacturing the cathode for an all-solid-state battery according to the present disclosure and the formation of the cathode through individual steps.

(18) Individual steps are described in detail with reference to FIG. 1.

(19) Preparation (S1)

(20) A protective member **10**, including a protective layer **12** and a mask layer **11** disposed on the protective layer **12** and having therein central portions **20**, which are columnar empty spaces, is prepared. More specifically, the protective member **10** is configured such that the protective layer **12** and the mask layer **11** are stacked in contact with each other and the protective layer **12** and the mask layer **11** have holes that are formed in a predetermined pattern. Here, the pattern is folioed of a plurality of circular or polygonal shapes, but this is merely an option that may be selected depending on the purpose of the invention, and the present disclosure is not limited thereto.

(21) The mask layer **11** needs to be folioed of a material that is resistant to a solvent or the like contained in the cathode material **40**. In particular, in the all-solid-state battery, a nonpolar solvent such as xylene or heptane may be used, and thus a material resistant thereto may be used.

(22) Specifically, the mask layer **11** includes polyethylene naphthalate (PEN).

(23) The protective layer **12** needs to be formed of a material that is resistant to a solvent or the like contained in the cathode material **40**. When drying the electrode, the material should be capable of withstanding a certain temperature (e.g. about 120° C.) and preferably has a coefficient of thermal expansion similar to that of the cathode current collector **30**.

(24) Specifically, the protective layer **12** includes polyethylene naphthalate (PEN).

(25) The cathode current collector **30** functions to transport electrons generated from a cathode coating layer **40'**, and may include any one selected from the group consisting of nickel, copper, zinc, aluminum and combinations thereof.

(26) The mask layer **11** is stacked on the upper surface of the protective layer **12**, and a release sheet may be attached to the lower surface of the protective layer **12**. Here, the release sheet may be removed before attachment of the protective member **10** to the upper surface of the cathode current collector **30**.

(27) The protective layer **12** and the mask layer **11** are stacked in contact with each other, and an adhesive may be interposed between the protective layer **12** and the mask layer **11**.

(28) The adhesive is used to prevent distortion of the protective layer **12** and the mask layer **11**, which are laminated with each other, by appropriately selecting the adhesiveness thereof, or to prevent the cathode material **40** from penetrating between the mask layer **11** and the protective layer **12** when coating with the cathode material **40**. Moreover, it is used to facilitate the removal of the mask layer **11** from the protective layer **12** without using a large force.

(29) The adhesive may include a silicone-based pressure sensitive adhesive (PSA) that is easily removable.

(30) The PSA includes acrylate or silicone, and may further include any one selected from the group consisting of ester rubber, phenol resin, castor oil, polyisobutylene and combinations thereof.

(31) The adhesive of the present disclosure, including silicone, has superior heat resistance, cold resistance, water resistance, insulating properties, ozone resistance, low combustibility, chemical inertness and non-pollutiveness, does not melt at high temperatures, and exhibits flexibility and adhesiveness even at very low temperatures.

(32) In order to enhance adhesion, a silicone adhesive composition including any one functional group selected from the group consisting of a hydroxyl group, an alkoxy group, an epoxy group and combinations thereof may be used as the adhesive.

(33) Stacking (S2)

(34) The protective member **10** is stacked on the cathode current collector **30**. Here, stacking the protective member **10** on the cathode current collector **30** may be performed using either or both of heat and pressure.

(35) Moreover, an adhesive layer may be interposed between the cathode current collector **30** and the protective member **10**. More specifically, an adhesive layer may be interposed between the cathode current collector **30** and the protective layer **12** of the protective member **10**.

(36) The adhesive layer needs to be folioed of a material that is resistant to a solvent or the like contained in the cathode material **40** and should be capable of withstanding an electrode drying temperature (e.g. about 120° C.) Moreover, a coefficient of thermal expansion may be similar to that of the cathode current collector **30**. The adhesive layer is different from the adhesive of the present disclosure in that the protective layer **12** and the cathode current collector **30** are attached such that they are not detached from each other.

(37) The adhesive layer may include a thermoplastic adhesive, and specifically, the thermoplastic adhesive includes any one selected from the group consisting of ethylene vinyl acetate (EVA) copolymer, polyvinyl acetate (PVA), polyethylene (PE), amorphous polypropylene, thermoplastic elastomer, polyamide, polyester and combinations thereof.

(38) The application of the thermoplastic adhesive described above enables continuous manufacture and the formation of an adhesive layer having a uniform adhesion surface.

(39) Coating (S3)

(40) The protective member **10** is coated with the cathode material **40** so that the central portions **20** of the protective member **10** are filled with the cathode material **40**. Specifically, the cathode material **40** is applied on the central portions **20** of the protective member **10**, which are empty spaces forming a predetermined pattern, and on the mask layer **11** of the protective member **10**, whereby the central portions **20** are filled with the cathode material **40**. Although the mask layer **11** may be fully covered with the cathode material **40**, only the central portions **20**, rather than the mask layer **11**, may be filled as necessary.

(41) The cathode material includes an active material, a binder and a solvent, and may further

include a conductor for increasing electronic conductivity as necessary. Here, the binder, the solvent and the conductor may be materials typically used in conventional secondary batteries and the like, and are not particularly limited in the present disclosure.

(42) It is sufficient that the active material can be used in the technical field of all-solid-state batteries, for example, nickel-cobalt-manganese (NCM), nickel-cobalt-aluminum (NCA), lithium-manganese oxide (LMO), lithium-cobalt oxide (LCO), nickel-manganese oxide (NMO), etc. may be included in the active material.

(43) After the coating process, drying the electrode material may be further performed.

(44) Through the drying process, the solvent is removed from the electrode material and the density of the active material and the binder contained in the electrode material may be further increased.

(45) Removal (S4)

(46) A cathode coating layer **40'** is formed by removing the mask layer **11** from the protective member **10**. Specifically, when the cathode material **40** is applied on the mask layer **11**, the cathode material **40** applied on the mask layer **11** is removed together with the mask layer **11**. Here, only the cathode coating layer **40'**, dried in the form of being inserted into the central portions **20**, is left behind. More specifically, the cathode coating layer **40'** is left behind in the form of a column having the pattern shape of the central portions **20**. Hence, the mask layer **11** is removed, whereby the thickness of the cathode coating layer **40'** included in the central portions **20** is naturally greater than the thickness of the protective layer **12**.

(47) After the removal process, the cathode coating layer **40'** is pressurized, thereby further increasing the density of the active material and the binder included in the cathode coating layer **40'**. Here, the thickness of the cathode coating layer **40'** may become equal to or smaller than the thickness of the protective layer **12**. FIG. 2 shows changes in the thickness of the cathode material **40** after the coating process, the thickness of the cathode coating layer **40'** after the removal process and the thicknesses of the protective layer **12** and the mask layer **11**. As shown in FIG. 2, the thickness L3 of the cathode material **40** is greater than the sum of the thickness L1 of the protective layer **12** and the thickness L2 of the mask layer **11**, and after the mask layer **11** is removed, the thickness L3' of the cathode coating layer **40'** is greater than the thickness L1 of the protective layer **12**. Subsequently, when pressure is applied to the cathode coating layer **40'**, the thickness L3' of the cathode coating layer **40'** may become equal to the thickness L1 of the protective layer **12**.

(48) According to the present disclosure, the thickness L1 of the protective layer **12** is 0.80 to 1.30 times the thickness L3' of the cathode coating layer **40'**, and preferably 0.85 to 1.25 times thereof. If the thickness is less than 0.80 times or is greater than 1.30 times thereof, the stack of the electrolyte and the anode formed on the cathode including the cathode coating layer **40'** may not be level, and there is a risk of internal short-circuiting.

(49) In the present disclosure, a punching process may be further performed after the removal process. The punching process is carried out to form a predetermined shape of the protective layer **12**, the cathode coating layer **40'** and the cathode current collector **30**, which are laminated together.

(50) The punching process is performed at a predetermined interval apart from the central portion **20** of the protective layer **12**, and the upper and lower surfaces of the punched protective layer **12** have a circular shape or a polygonal shape.

(51) FIG. 3 shows the punching process in which the upper and lower surfaces of the protective layer **12** have a circular shape according to an embodiment of the present disclosure. With reference thereto, the cathode coating layer **40'** is provided in the form of being inserted into the multiple central portions **20** of the protective layer **12**, and the punching process is performed in a circular shape at a predetermined interval apart from each of the multiple central portions **20** of the protective layer **12**. The punching process enables the outer appearance of the protective layer **12** and the cathode current collector **30** to be circular.

(52) An all-solid-state battery is manufactured from each of the multiple central portions **20** of the protective layer **12**, whereby a cathode including the protective layer **12** may be mass-produced

through a single process.

(53) Cathode for all-Solid-State Battery

(54) According to the present disclosure, the cathode for an all-solid-state battery includes a cathode current collector **30**, a cathode coating layer **40'** having a columnar shape formed on the cathode current collector **30**, and an insulating layer **12'** disposed on the cathode current collector **30** and having therein a central portion **20**, which is a columnar empty space.

(55) Specifically, the upper surface of the central portion **20** has a circular shape, and the cathode coating layer **40'** is inserted into the central portion **20**, and thus a portion of the outer side surface of the cathode coating layer **40'** and the side surface of the central portion **20** are in contact with each other.

(56) Here, the insulating layer **12'** is formed of the same type of material as the protective layer **12** of the present disclosure, and the insulating layer **12'** is configured to include only one central portion **20**, unlike the protective layer **12** in a plate form having multiple central portions **20** in the manufacturing method. The insulating layer **12'** may be obtained by punching the protective layer **12**, and there is no difference in material therebetween.

(57) Referring to FIG. 3 again, the protective layer **12** before the punching process is illustrated on the left side, and the insulating layer **12'** after the punching process is illustrated on the right side.

(58) The insulating layer **12'** may include an insulator. The insulating layer **12'** is used to prevent short-circuiting between electrodes or between the electrode and the electrolyte. As the insulator, any material may be used without particular limitation, so long as it is able to interrupt the flow of electrons.

(59) The thickness of the insulating layer **12'** may be smaller than or equal to the thickness of the cathode coating layer **40'**, as shown in FIG. 2.

(60) The thickness of the insulating layer **12'** is 100 μm or less, and the thickness of the cathode coating layer **40'** is 80 to 115 μm .

(61) FIGS. 4A to 4C show the outer diameter and the inner diameter of the insulating layer **12'** having a circular shape, and the diameter of the cathode coating layer **40'**, according to an embodiment of the present disclosure. With reference thereto, FIG. 4A shows the inner diameter $\varnothing.\text{sub.x}$ of the insulating layer **12'** and the outer diameter $\varnothing.\text{sub.y}$ of the insulating layer **12'**. The inner diameter $\varnothing.\text{sub.x}$ is equal to the diameter of the central portion **20** of the insulating layer **12'**, and the outer diameter $\varnothing.\text{sub.y}$ is equal to the diameter of the pattern that is punched.

(62) FIG. 4B shows the diameter $\varnothing.\text{sub.z}$ of the cathode coating layer **40'** formed in the central portion **20** of the insulating layer **12'**, in which the cathode coating layer **40'** is fully loaded in and inserted into the central portion **20** of the insulating layer **12'**. Hence, the diameter $\varnothing.\text{sub.z}$ of the cathode coating layer **40'** is equal to the diameter of the central portion **20**. Specifically, the diameter $\varnothing.\text{sub.z}$ of the cathode coating layer **40'** is equal to the inner diameter $\varnothing.\text{sub.x}$ of the insulating layer **12'**.

(63) In the present disclosure, the outer diameter $\varnothing.\text{sub.y}$ of the insulating layer **12'** satisfies Expression 1 below.

$$1.1\varnothing.\text{sub.x} \leq \varnothing.\text{sub.y} \leq 2.0\varnothing.\text{sub.x} \quad [\text{Expression 1}]$$

(64) Preferably, the outer diameter $\varnothing.\text{sub.y}$ of the insulating layer **12'** satisfies Expression 2 below.

$$1.1\varnothing.\text{sub.x} \leq \varnothing.\text{sub.y} \leq 1.5\varnothing.\text{sub.x} \quad [\text{Expression 2}]$$

(65) Here, if the outer diameter of the insulating layer **12'** is less than $1.1\varnothing.\text{sub.x}$, the area of the insulating layer **12'** is excessively small, and thus the insulating layer cannot function as an insulator for preventing short-circuiting between electrodes. On the other hand, if the outer diameter thereof exceeds $2.0\varnothing.\text{sub.x}$, the area of the cathode coating layer **40'** may decrease, thus deteriorating battery performance and efficiency.

(66) In the present disclosure, when the insulating layer **12'** has a polygonal shape, rather than a circular shape, the central portion has a polygonal shape corresponding to the shape of the insulating layer **12'**. Here, $\varnothing.\text{sub.y}$ designates the outer diameter of the insulating layer having a

polygonal shape, and $\varnothing_{\text{sub.x}}$ designates the inner diameter of the insulating layer in the direction in which the outer diameter is measured, which satisfies Expression 3 below.

$$0.1 \leq (\varnothing_{\text{sub.y}} - \varnothing_{\text{sub.x}}) \leq 1.0 \quad [\text{Expression 3}]$$

(67) When the insulating layer **12'** has a polygonal shape, the shape of the cathode coating layer inserted into the central portion may be polygonal depending on the above polygonal shape.

(68) All-Solid-State Battery

(69) According to the present disclosure, the all-solid-state battery includes the cathode for an all-solid-state battery of the present disclosure, an anode **60**, and a solid electrolyte **50** interposed between the cathode and the anode **60**.

(70) FIG. **4C** shows the cathode of the present disclosure including the cathode current collector **30**, the insulating layer **12'** and the cathode coating layer **40'**, with the solid electrolyte **50** and the anode **60** stacked on the cathode.

(71) The present disclosure pertains to the cathode for a battery including a solid electrolyte **50**, and does not particularly limit the types of the anode **60** or the solid electrolyte **50**.

(72) For example, LISICON-based compounds, thio-LISICON-based compounds, lithium sulfide ($\text{Li}_{\text{sub.2S}}$)-based compounds, lithium sulfide ($\text{Li}_{\text{sub.2S}}$)-phosphorus disulfide ($\text{P}_{\text{sub.2S.sub.5}}$) mixtures, aziridine-based compounds and the like can be used as the solid electrolyte **50**.

(73) Diameters of the anode **60** and the solid electrolyte **50** are equal to each other, the diameter $\varnothing_{\text{sub.w}}$ of the anode **60** and the solid electrolyte **50** needs to be greater than the diameter $\varnothing_{\text{sub.z}}$ of the cathode coating layer **40'** or the inner diameter $\varnothing_{\text{sub.x}}$ of the insulating layer **12'**, and smaller than the outer diameter $\varnothing_{\text{sub.y}}$ of the insulating layer **12'**. If the diameters fall out of the above ranges, there is a risk of short-circuiting between electrodes and battery efficiency may deteriorate.

(74) A better understanding of the present disclosure will be given through the following examples. However, these examples are merely set forth to illustrate the present disclosure, and are not to be construed as limiting the scope of the present disclosure.

Example 1

(75) A protective member, configured such that a protective layer (PEN film) having a thickness of about 90 μm and a mask layer (PEN film) having a thickness of 30 μm were sequentially attached to the upper surface of a release film, was punched to form a central portion in a circular pattern having a diameter of 14 μmm , the release film was removed, and the protective member was thermally fused to a cathode current collector (Al foil) having a polyethylene film attached thereto by applying heat at 100° C. Here, a silicone adhesive was interposed between the protective layer and the mask layer.

(76) On the mask layer and the cathode current collector exposed through the central portion, a cathode material including lithium cobalt oxide ($\text{LiCoO}_{\text{sub.2}}$), carbon black and polyvinylidene fluoride (PVdF) was applied and dried in a vacuum for 8 hr.

(77) Thereafter, the mask layer and the dried cathode material on the mask layer were removed, and pressure was applied using a press so that the average thickness of the cathode coating layer remaining in the form of being inserted into the central portion was 100 μm .

(78) The cathode current collector including the protective layer and the cathode coating layer was punched in a circular pattern having a diameter of 18 μmm .

(79) Thereafter, a solid electrolyte having a diameter of 16 mm and including lithium (Li), lanthanum (La), zirconium (Zr), oxide (O) and polyvinylidene fluoride, and an anode including graphite and polyvinylidene fluoride were sequentially stacked and pressed to respective average thicknesses of 40 μm and 90 μm , thereby manufacturing an all-solid-state battery (in which the amounts of the cathode, the anode and the electrolyte therein were 23.22 $\mu\text{mg/cm}^2$, 15.33 $\mu\text{mg/cm}^2$ and 6.18 $\mu\text{mg/cm}^2$, respectively).

(80) FIG. **5** shows the cross-sectional images of the all-solid-state battery of Example 1 manufactured above. With reference thereto, it can be confirmed that the cathode coating layer and

the insulating layer have almost the same thickness and thus the side surface of the cathode is completely protected by virtue of the insulating layer.

(81) FIG. 6 is a charge/discharge graph of the all-solid-state battery manufactured above, particularly a charge/discharge graph after 200 charge/discharge cycles at a voltage (V) ranging from 2.5 to 4.3. With reference thereto, it can be confirmed that stable charge/discharge results appear even when the solid electrolyte is thin.

Comparative Examples 1 to 3

(82) On a cathode current collector, a cathode material including lithium cobalt oxide (LiCoO₂), carbon black and polyvinylidene fluoride (PVdF) was applied and dried in a vacuum for 8 hr, and pressure was then applied thereto using a press so as to attain an average thickness of 100 μm. The cathode current collector coated with the cathode material was punched in a circular pattern having a diameter of 18 μm.

(83) Thereafter, the solid electrolyte and the anode, which were the same as in Example 1, were prepared and sequentially stacked, thus manufacturing three all-solid-state batteries.

(84) The three all-solid-state batteries, particularly those of Comparative Example 1 (E1), Comparative Example 2 (E2) and Comparative Example 3 (E3), were subjected to a charging test. The results are shown in FIG. 7. With reference thereto, it can be seen that these batteries do not operate like normal cells. This is deemed to be caused by short-circuiting between the cathode and the anode attributed to cracking of the solid electrolyte at the boundary portion upon pressurization of the cell due to the difference in area between the cathode and the anode.

(85) As is apparent from the results of Example 1 and Comparative Examples 1 to 3, the protective member including the protective layer and the mask layer is applied, whereby the patterned cathode coating layer and insulating layer can be efficiently aligned with each other. Moreover, the edge of the cathode coating layer is protected by the insulating layer, and thus, when the all-solid-state battery is pressurized, problems of cracking of the solid electrolyte in contact with the edge portion of the cathode coating layer and short-circuiting between electrodes are solved, ultimately enabling stable charging and discharging performance to be realized.

(86) Although the embodiments of the present disclosure have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications are possible without departing from the scope and spirit of the invention as disclosed in the accompanying claims, and such modifications should not be understood separately from the technical ideas or essential characteristics of the present disclosure.

Claims

1. A method of manufacturing a cathode for an all-solid-state battery, the method comprising: stacking a protective member on a cathode current collector, the protective member including a protective layer and a mask layer disposed on the protective layer and having a central portion which is an empty space; coating the protective member with a cathode material so that the central portion of the protective member is filled with the cathode material; removing the mask layer to form a cathode coating layer; and pressurizing the cathode coating layer, punching the protective layer, the cathode coating layer, and the cathode current collector after removing the mask layer, wherein an insulating layer is formed through the punching against the protective layer, wherein the cathode coating layer has a thickness of 80 to 115 μm, wherein an adhesive layer is interposed between the cathode current collector and the protective member, wherein the adhesive layer includes any one selected from the group consisting of an ethylene vinyl acetate (EVA) copolymer, polyvinyl acetate (PVA), polyethylene (PE), amorphous polypropylene, a thermoplastic elastomer, polyamide, polyester, and combinations thereof, wherein the insulating layer has a circular shape with an inner diameter $\phi_{sub.x}$ and an outer diameter $\phi_{sub.y}$, and wherein the inner diameter $\phi_{sub.x}$ and the outer diameter $\phi_{sub.y}$ of the insulating layer satisfy Expression 1 below:

1.1 $\varnothing$.sub.x $\leq$ 0.sub.y $\leq$ 2.0 $\varnothing$.sub.x. [Expression 1]

2. The method of claim 1, wherein the mask layer includes polyethylene naphthalate (PEN), and wherein the protective layer includes polyethylene naphthalate (PEN).
 3. The method of claim 1, wherein the mask layer is disposed on an upper surface of the protective layer.
 4. The method of claim 3, wherein an adhesive is interposed between the protective layer and the mask layer.
 5. The method of claim 1, wherein the protective member is stacked on the cathode current collector by applying at least one of heat or pressure.
 6. The method of claim 1, wherein, in the removing the mask layer, the mask layer and the cathode material applied on the mask layer are removed, and the cathode material is maintained in a shape of being inserted into the central portion of the protective layer as the cathode coating layer.
 7. The method of claim 1, wherein a thickness of the cathode material is greater than or equal to a sum of a thickness of the protective layer and a thickness of the mask layer.
 8. The method of claim 1, wherein the punching the protective layer, the cathode coating layer, and the cathode current collector is performed at a predetermined interval apart from the central portion of the protective layer, and wherein an upper surface and a lower surface of the punched protective layer have a circular shape.
 9. The method of claim 1, wherein pressurizing the cathode coating layer is performed after removing the mask layer.
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